

Lorentz Force

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$$\begin{aligned}\vec{F}_L &= \vec{F}_e + \vec{F}_m \\ \vec{F}_L &= q\vec{E} + q\vec{v} \times \vec{B}\end{aligned}$$

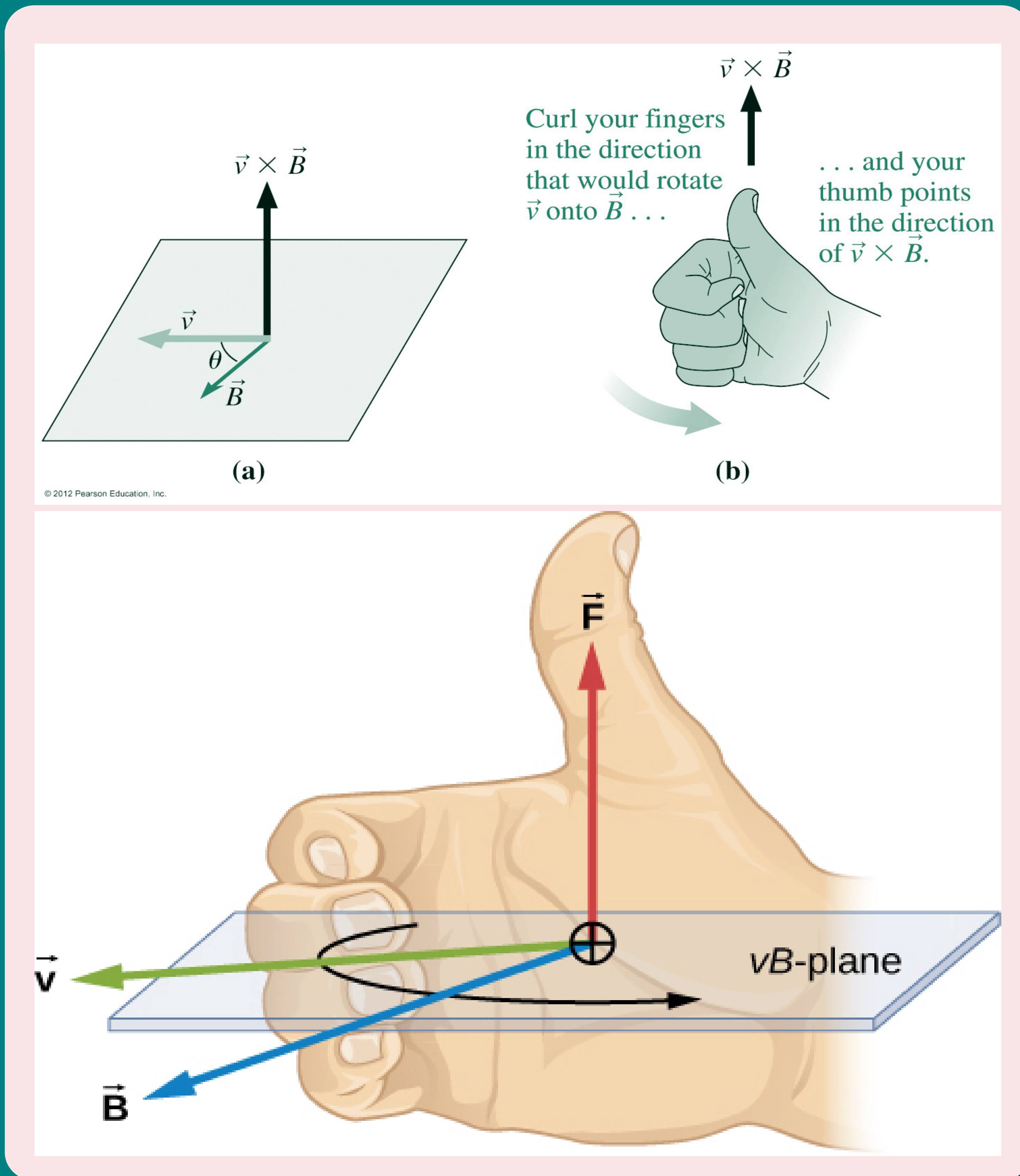
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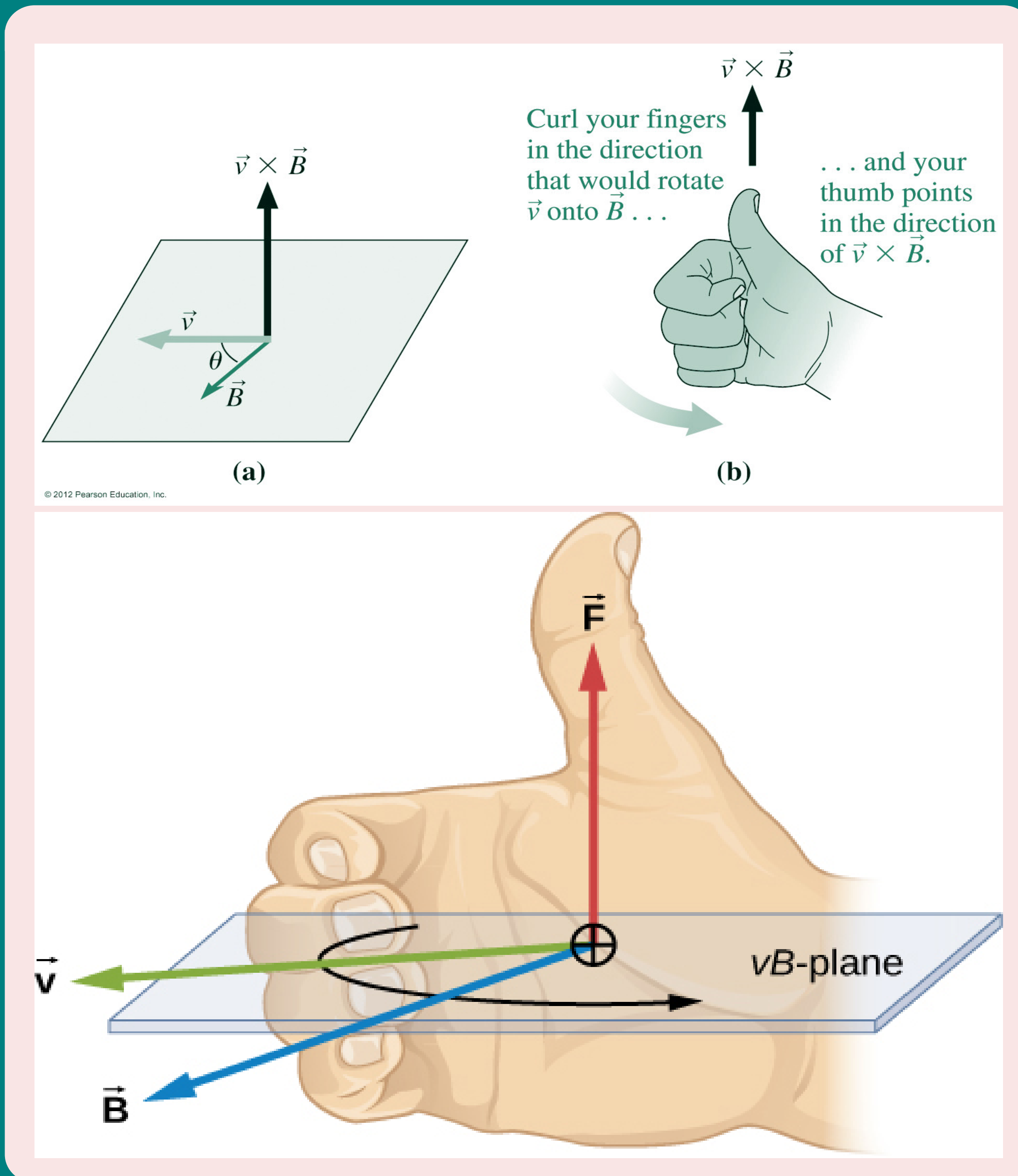
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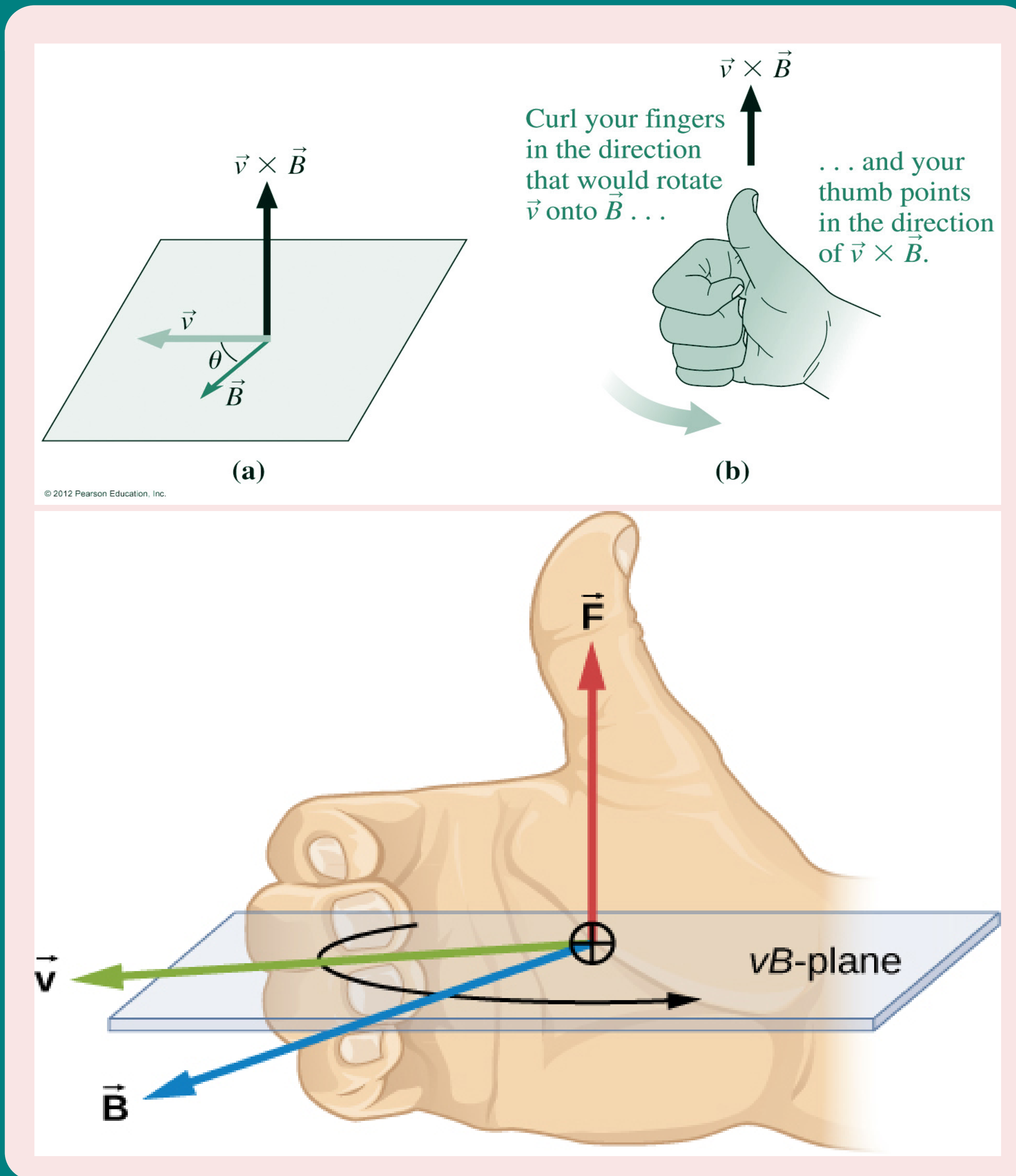
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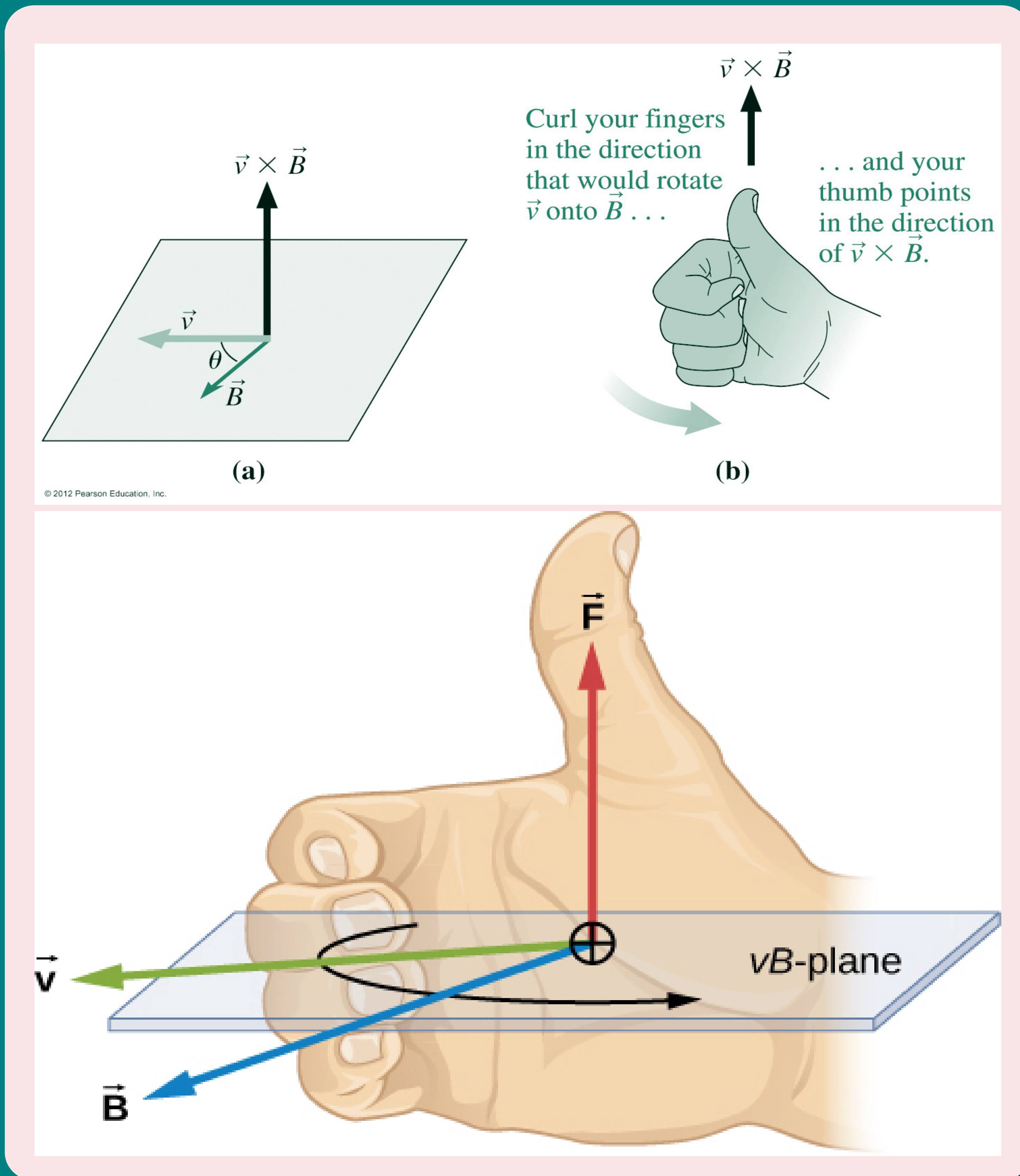
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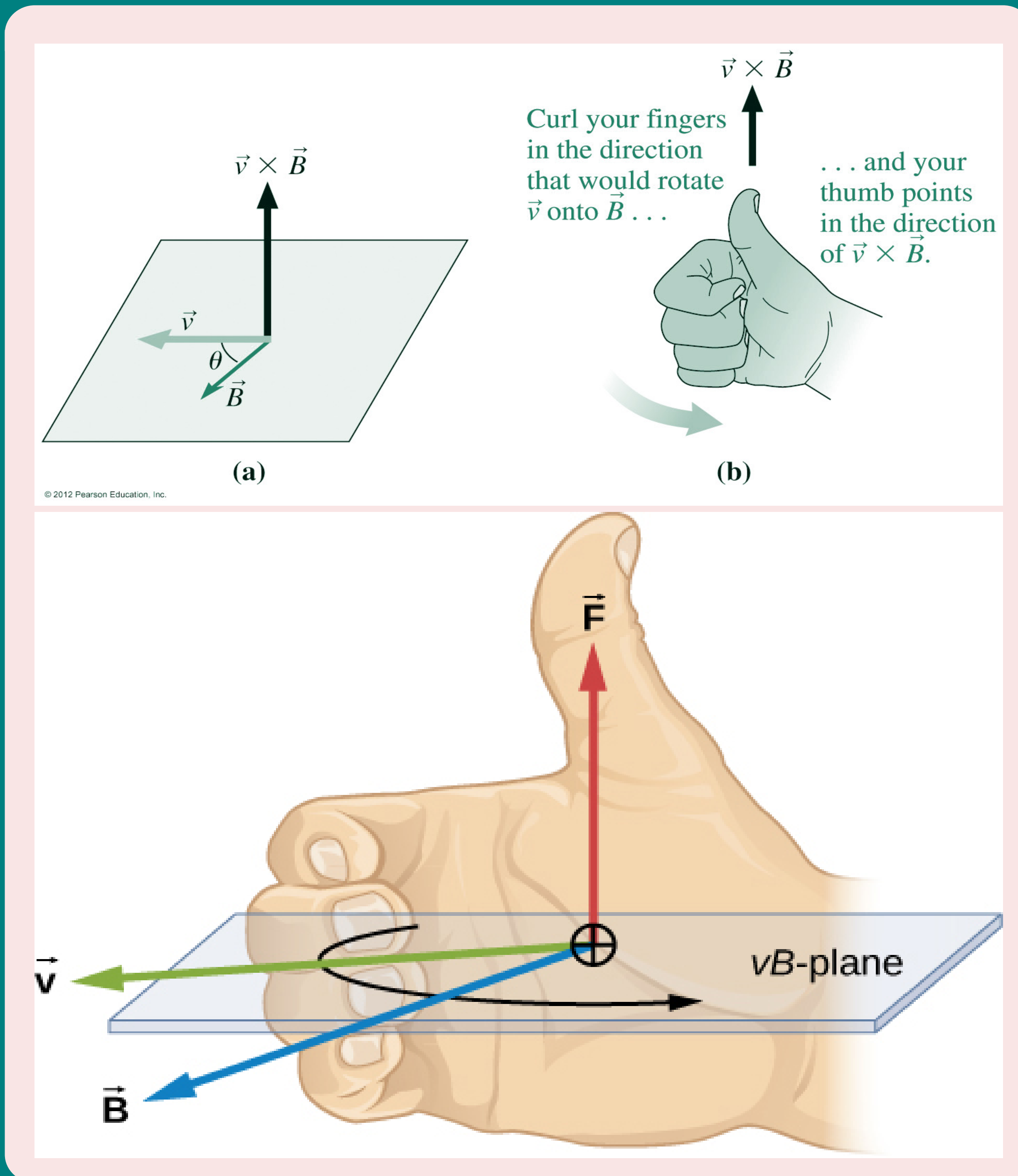
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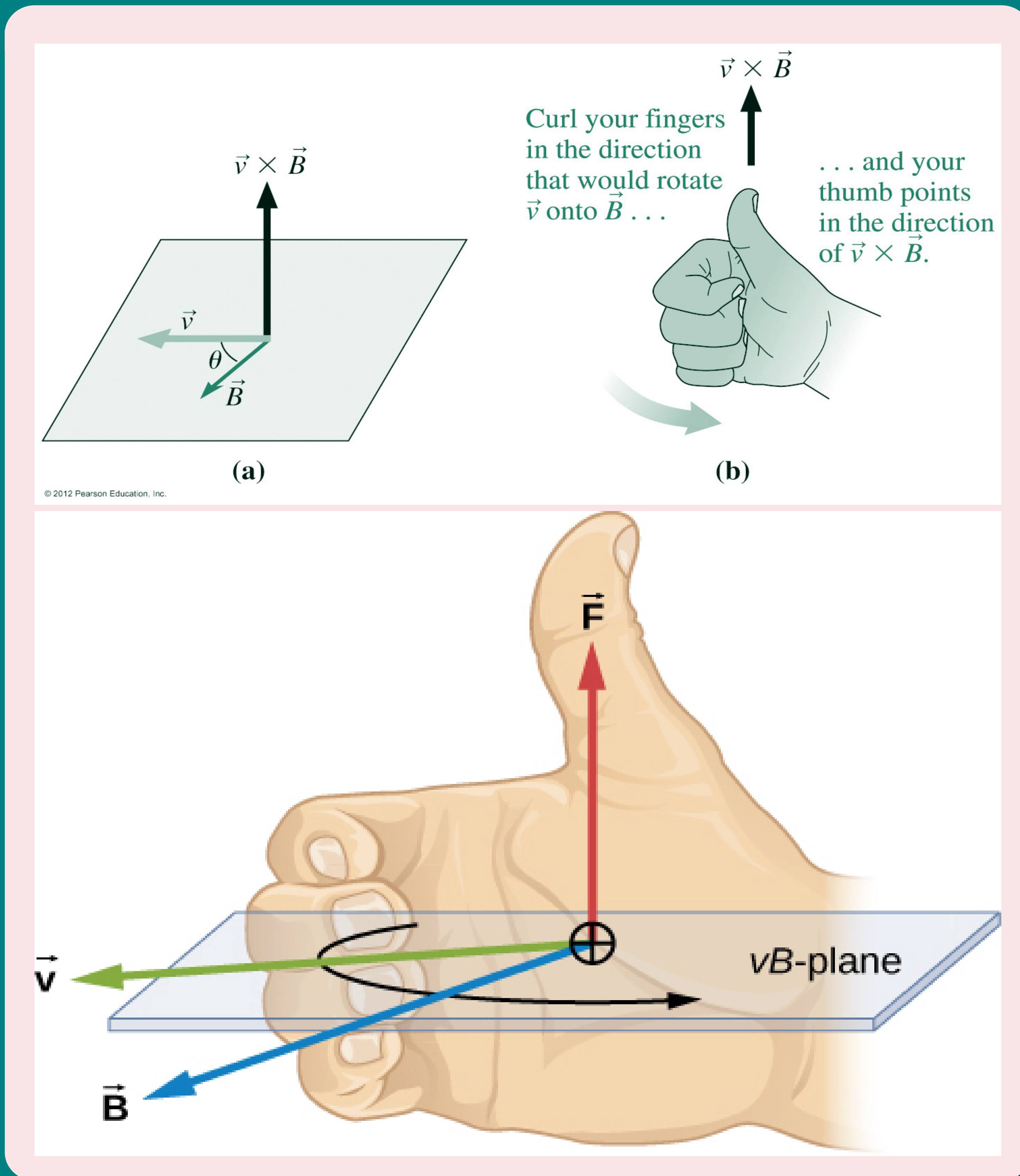
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- **Magnetic Lorentz force $\vec{F}_m$ will always perpendicular to both $\vec{v}$ and $\vec{B}$ and its direction is given by **Right hand thumb rule** for vector product.**

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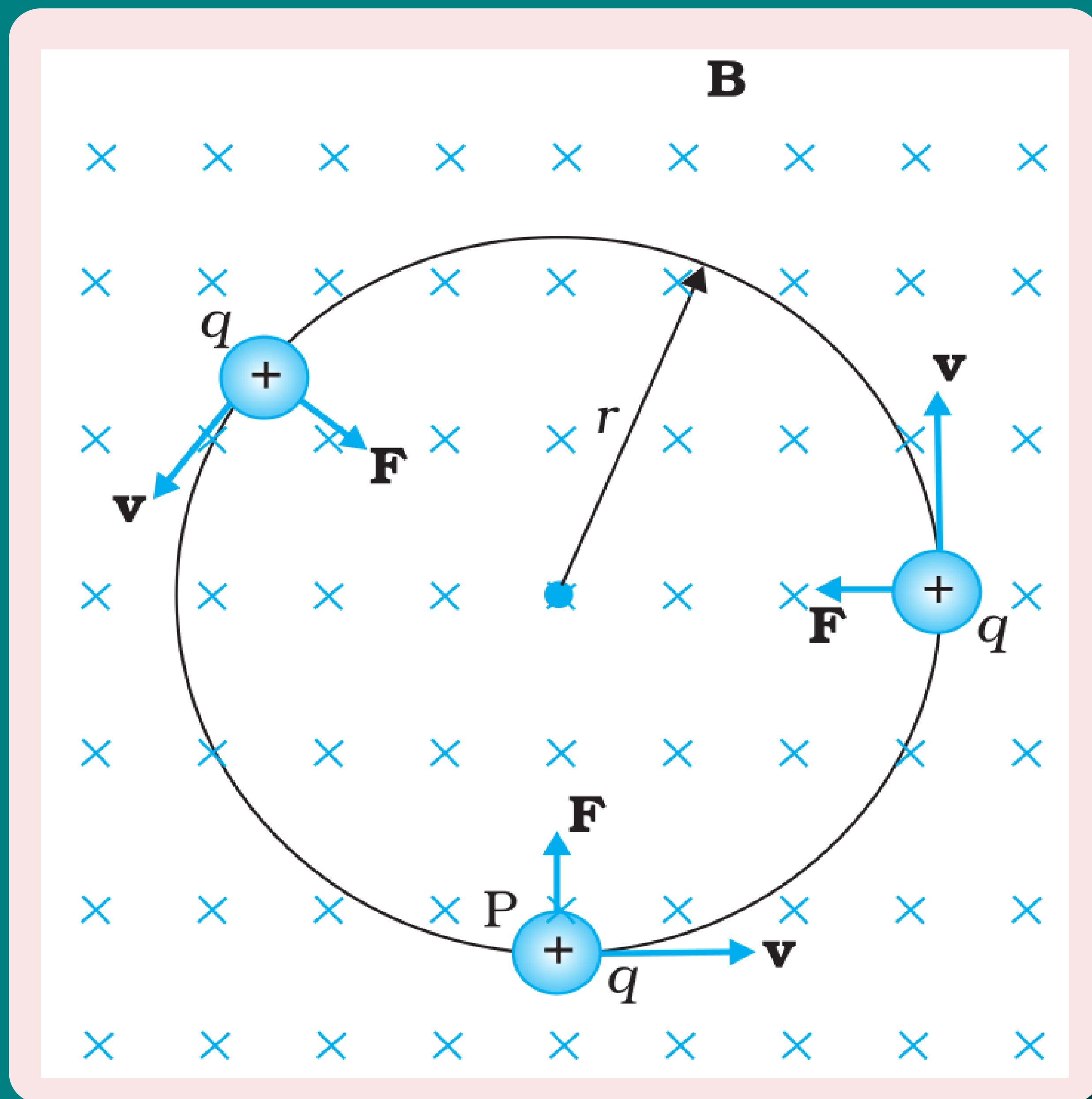
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$$1T = \frac{N \ s}{C \ m}$$

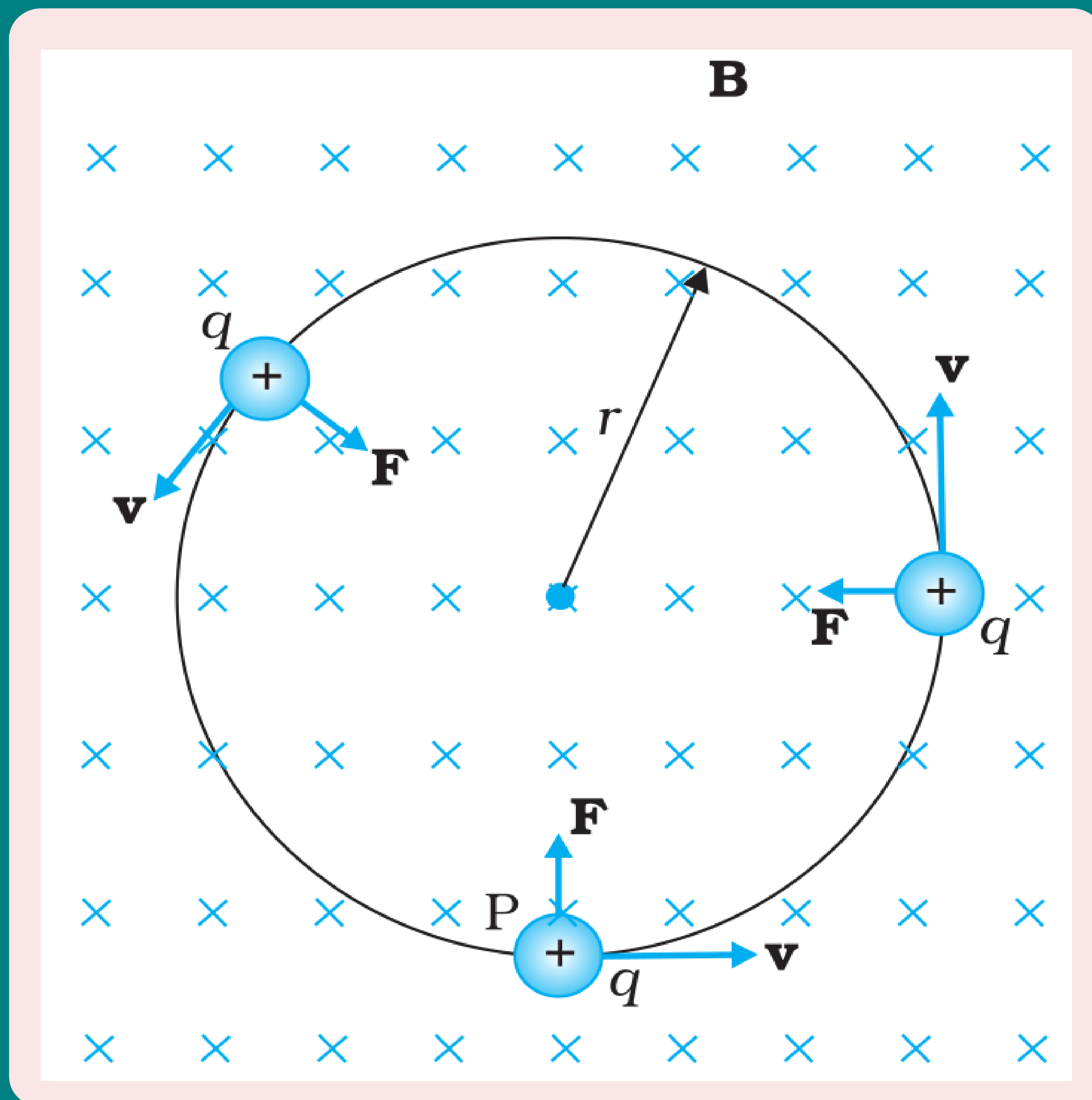
- **A smaller unit, called the gauss (G), where $1G=10^{-4} \ T$, is sometimes used.**

Motion of a Charged Particle in a Magnetic Field



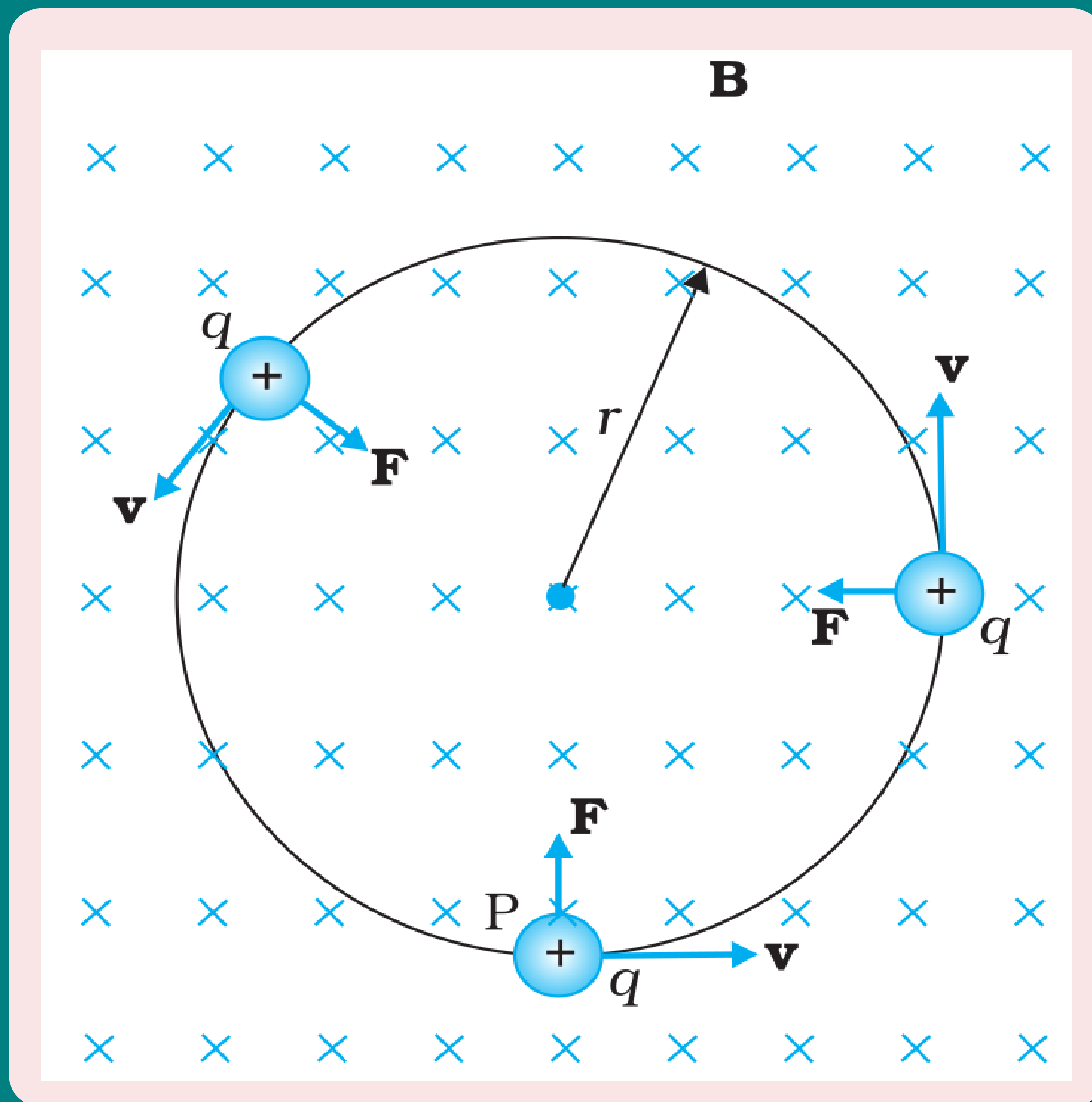
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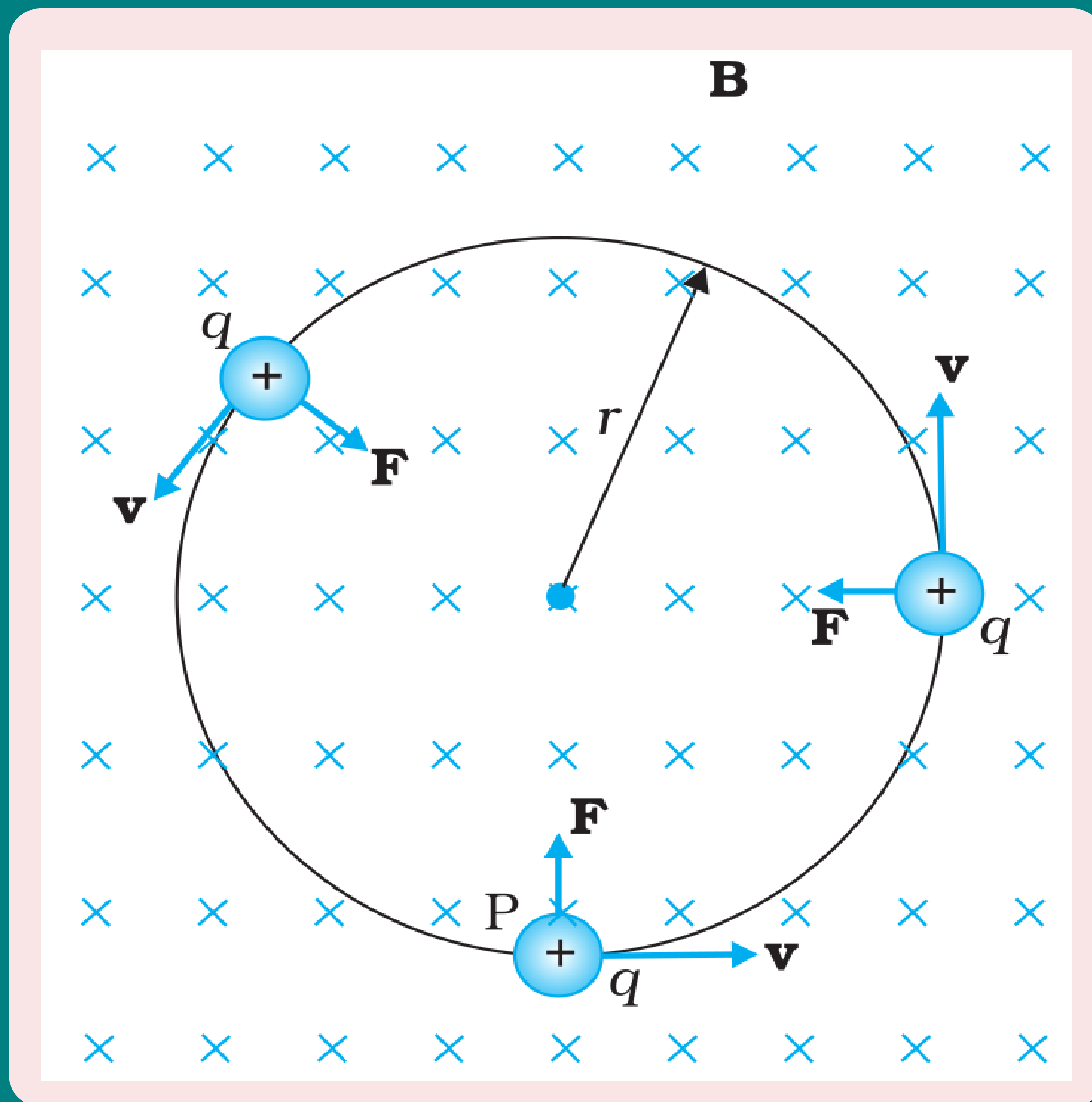
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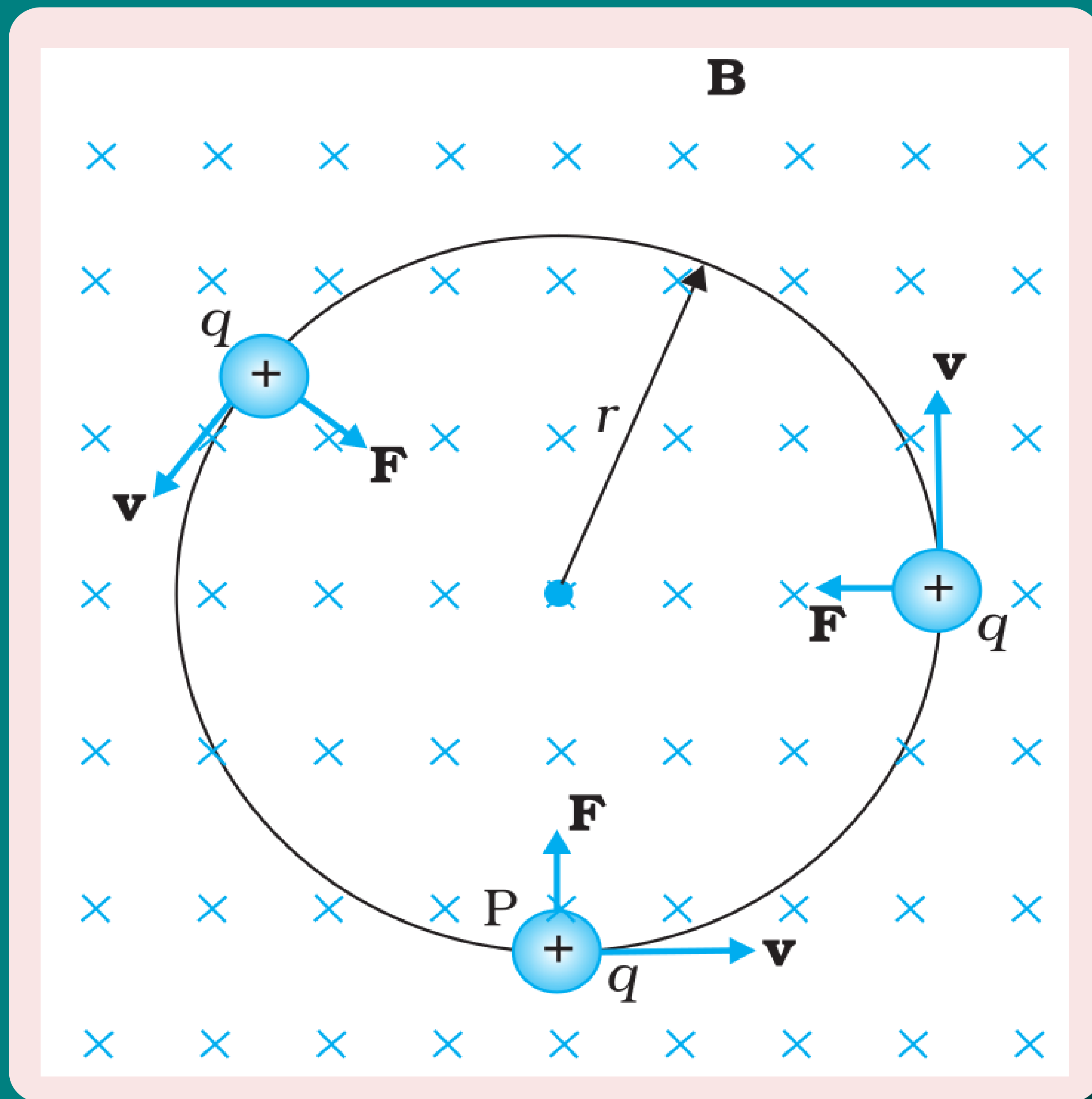
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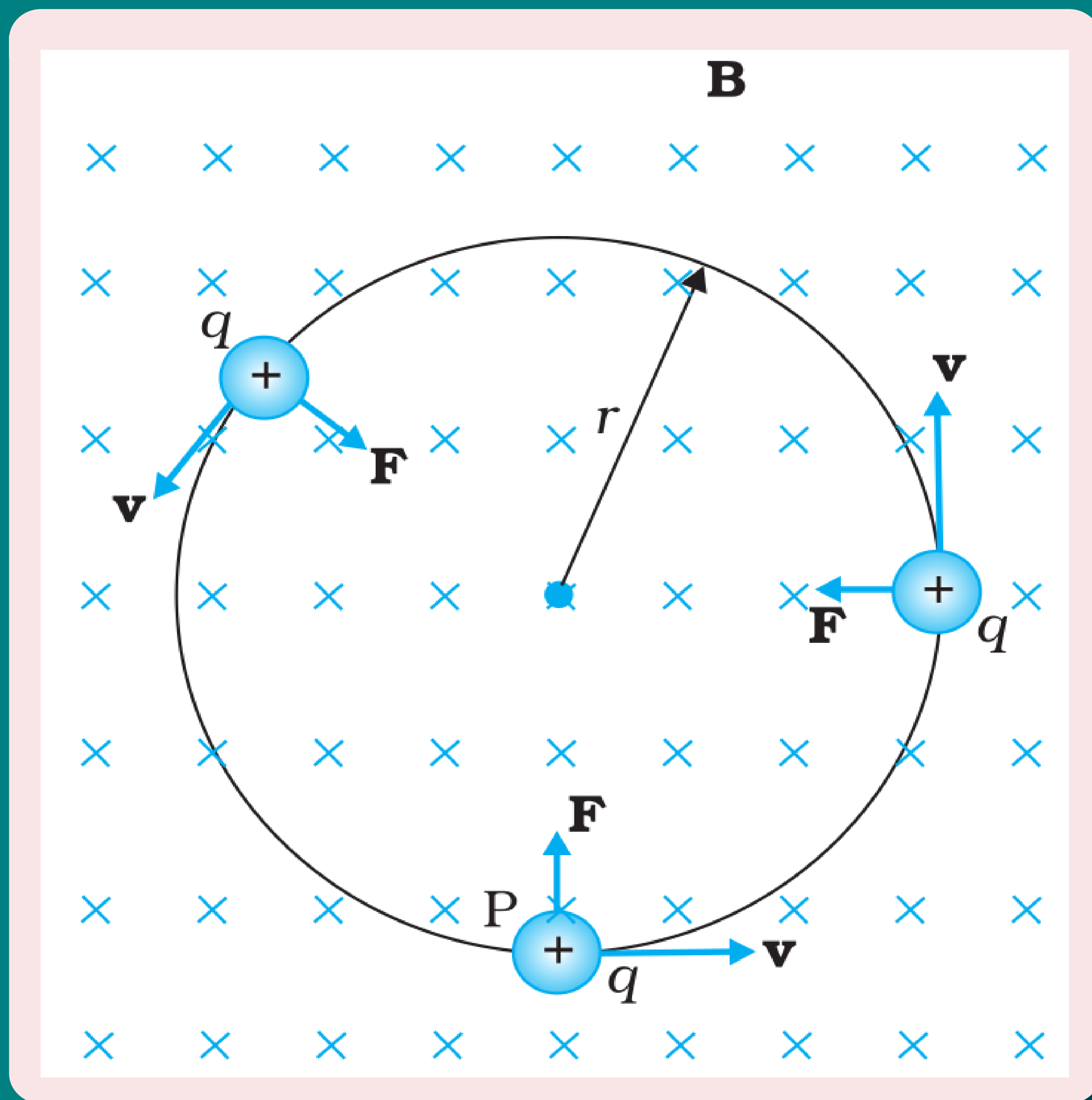
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- The **direction of motion is affected but not the speed**.

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Case 1: When $\vec{v}$ perpendicular to $\vec{B}$

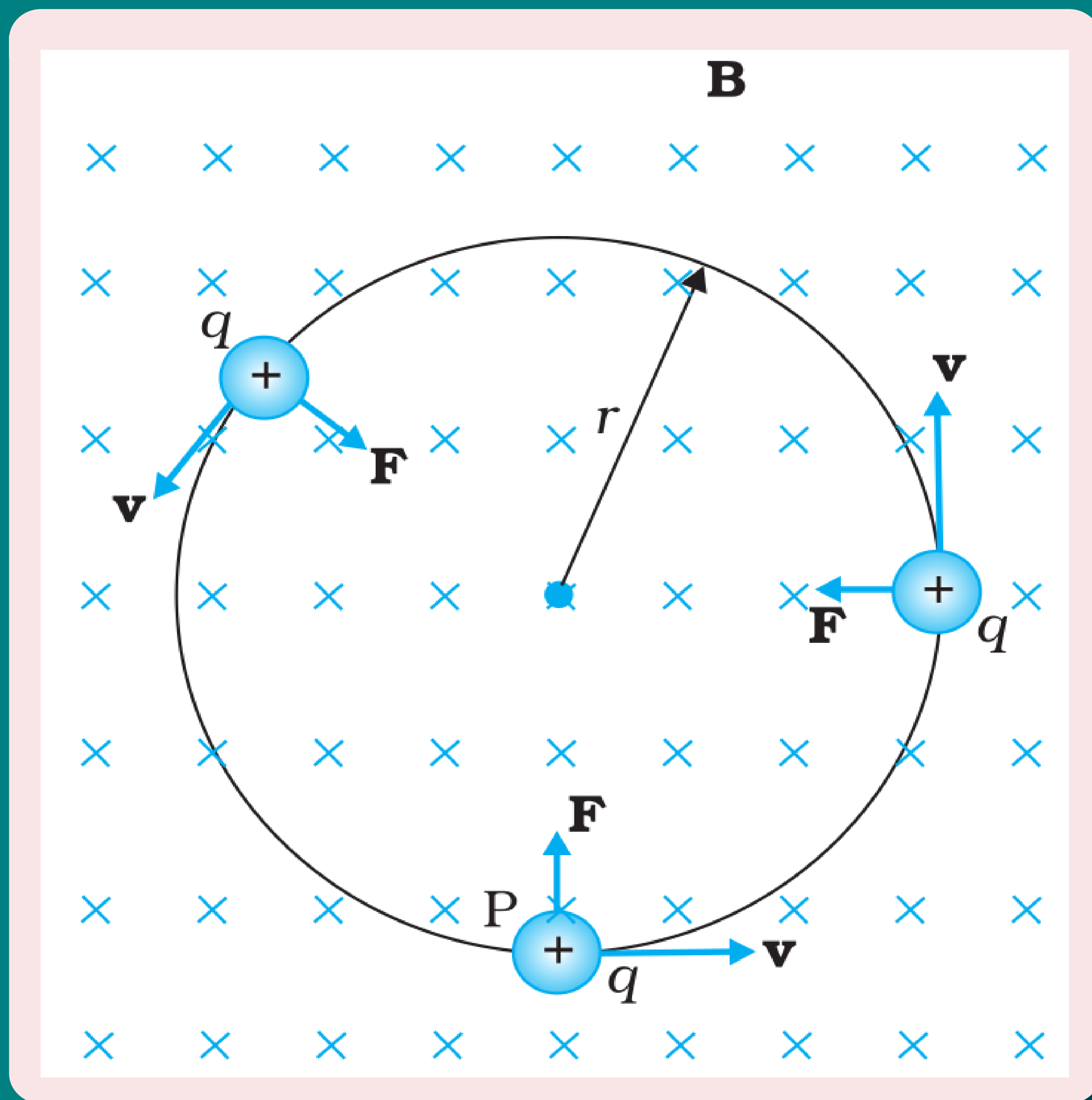
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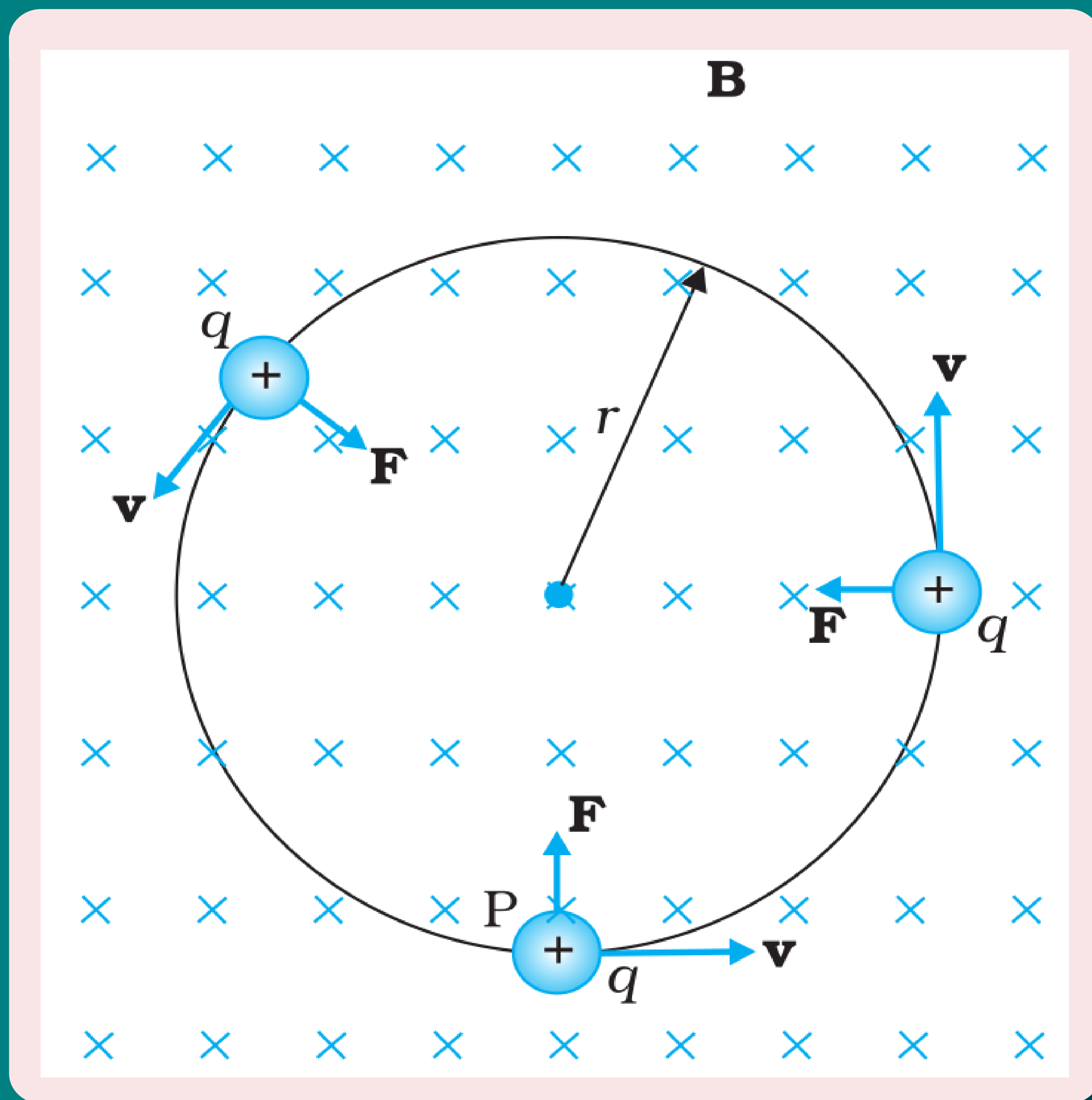


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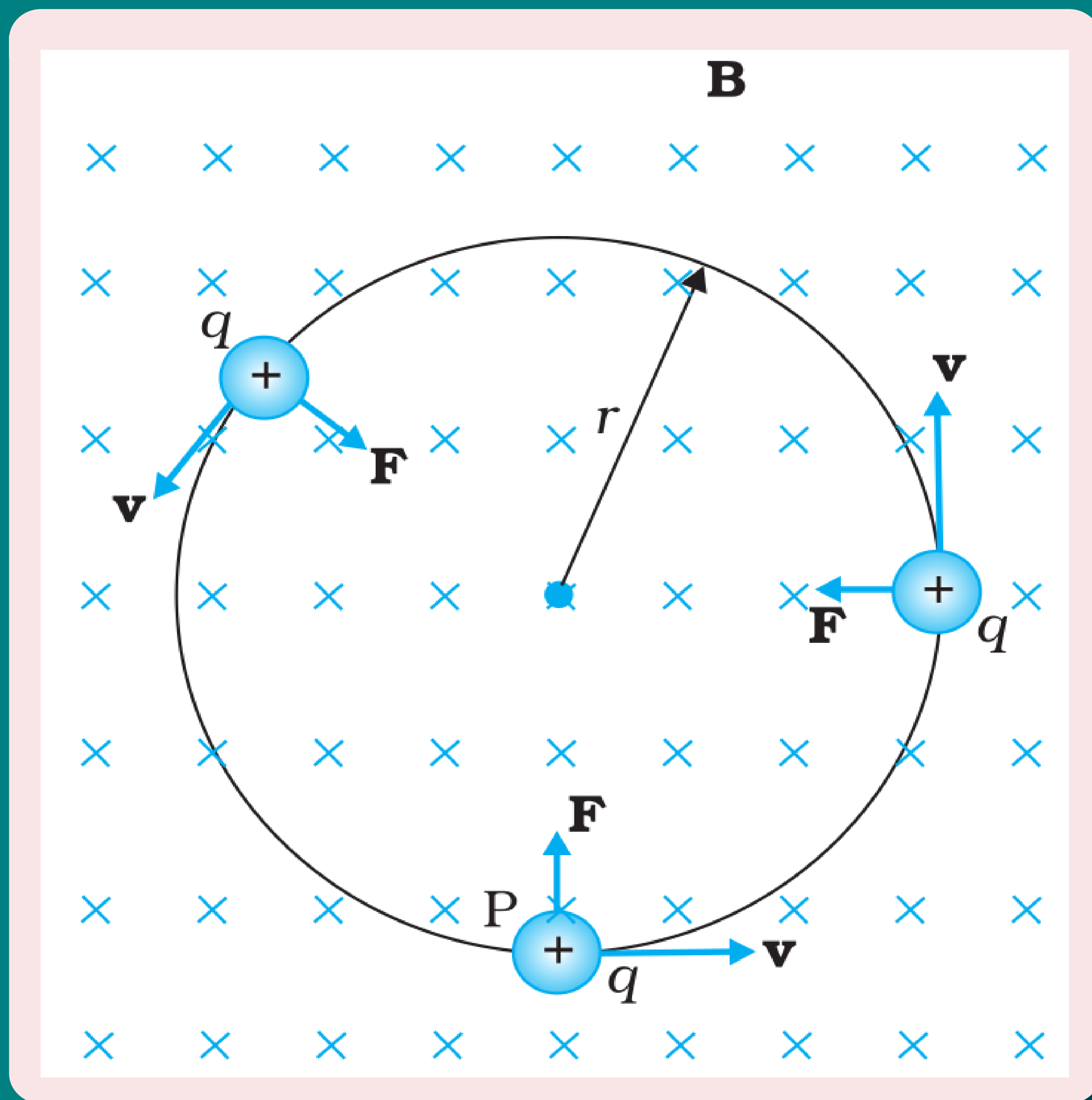
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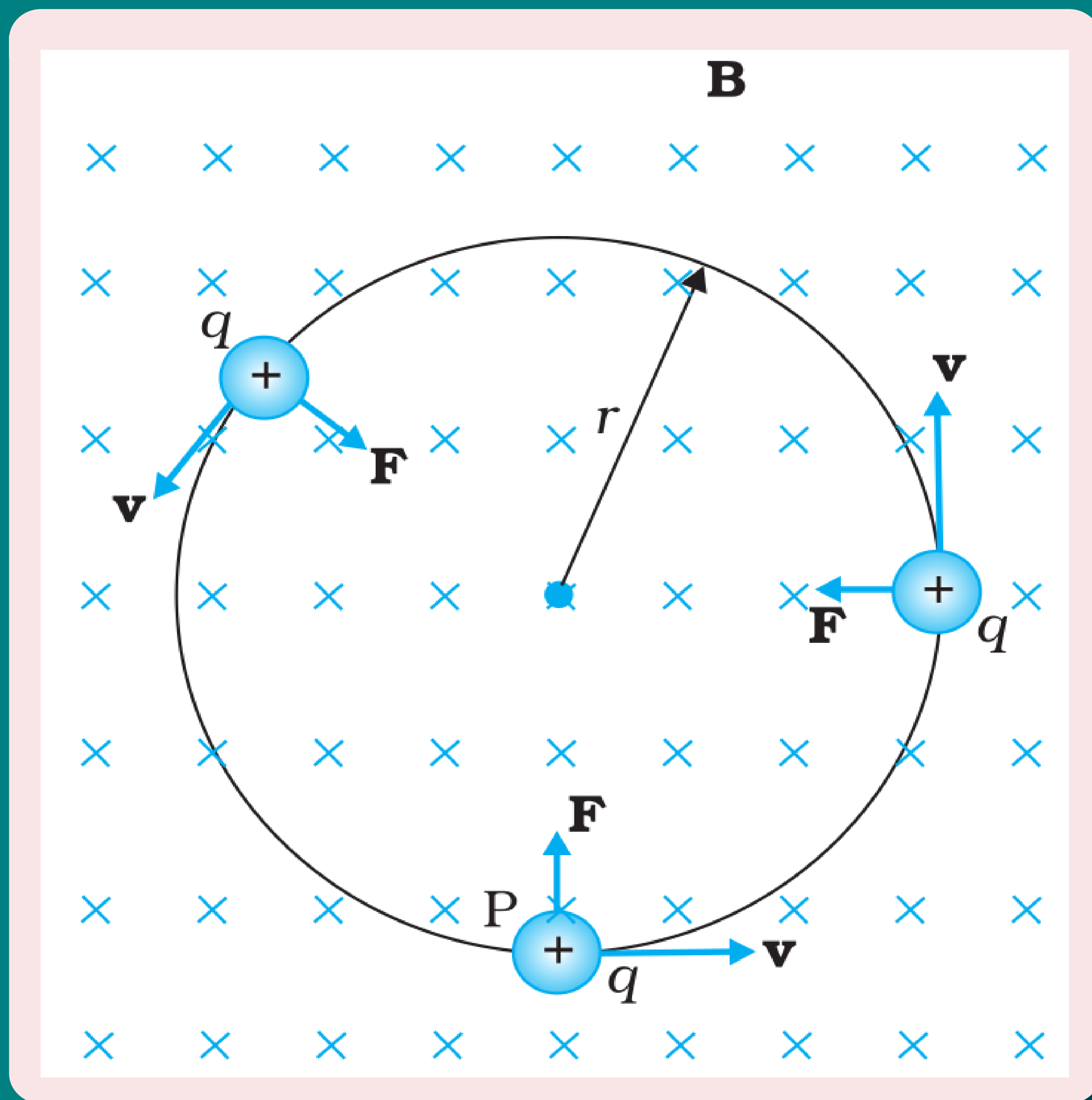
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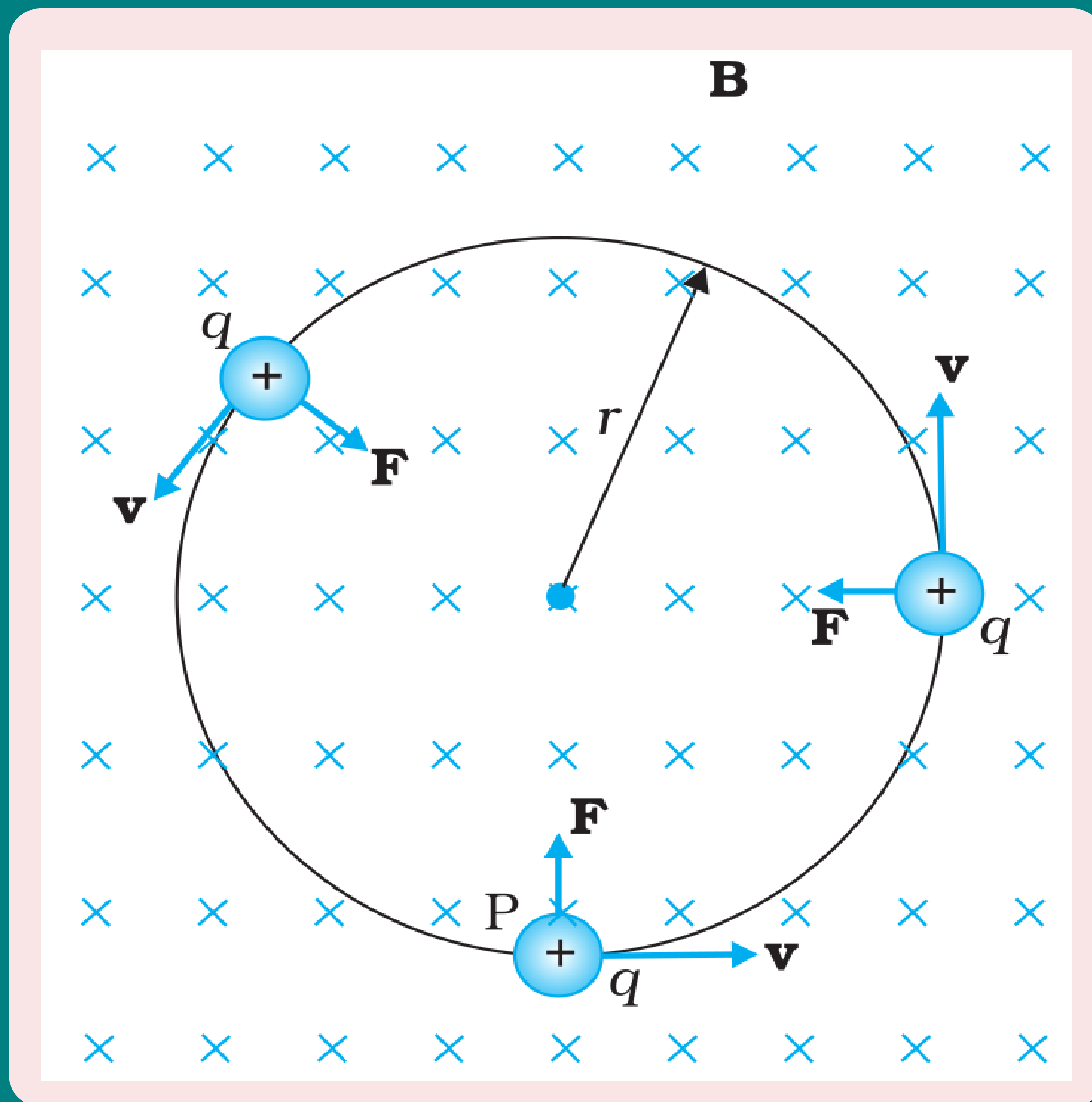
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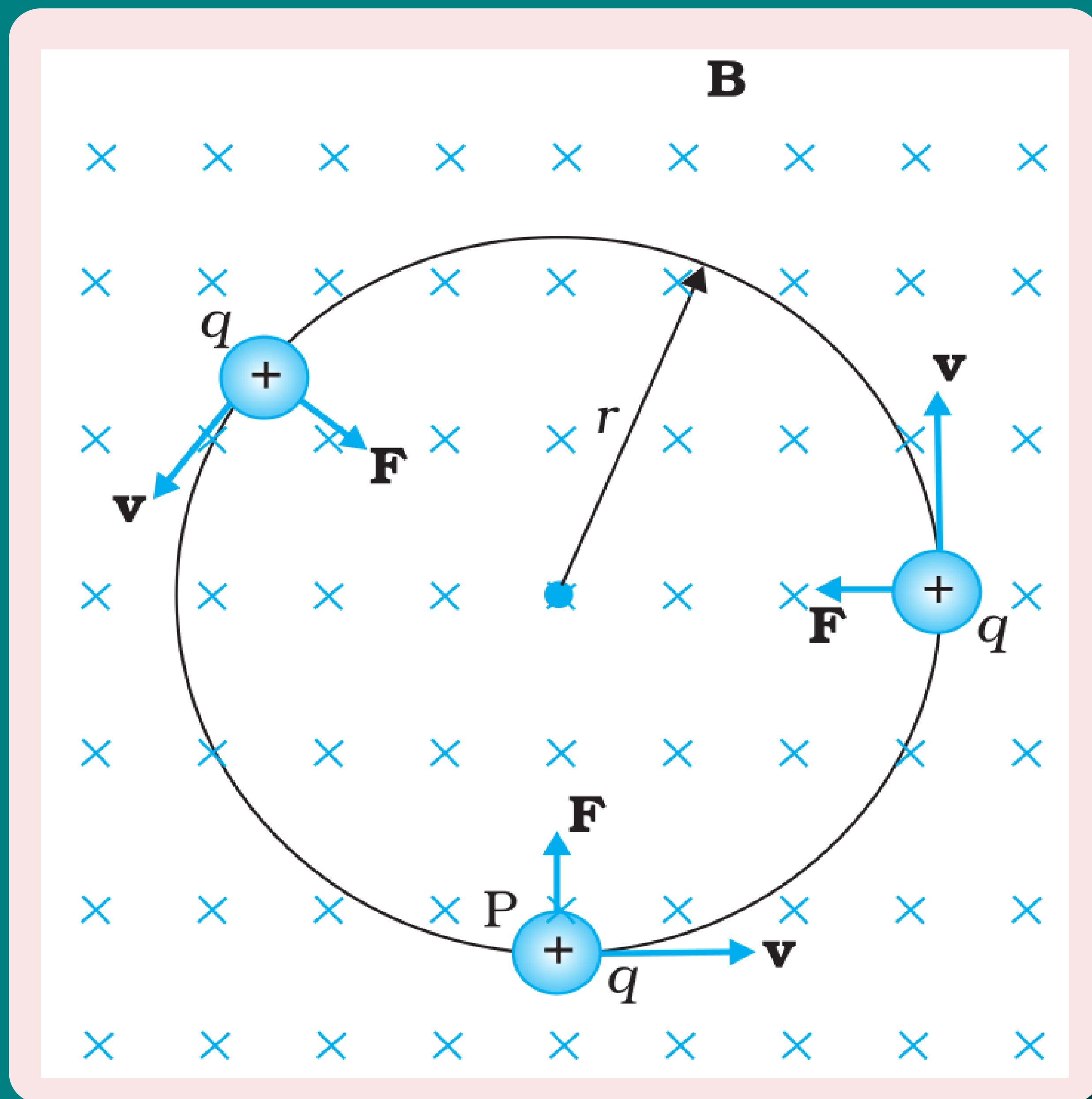
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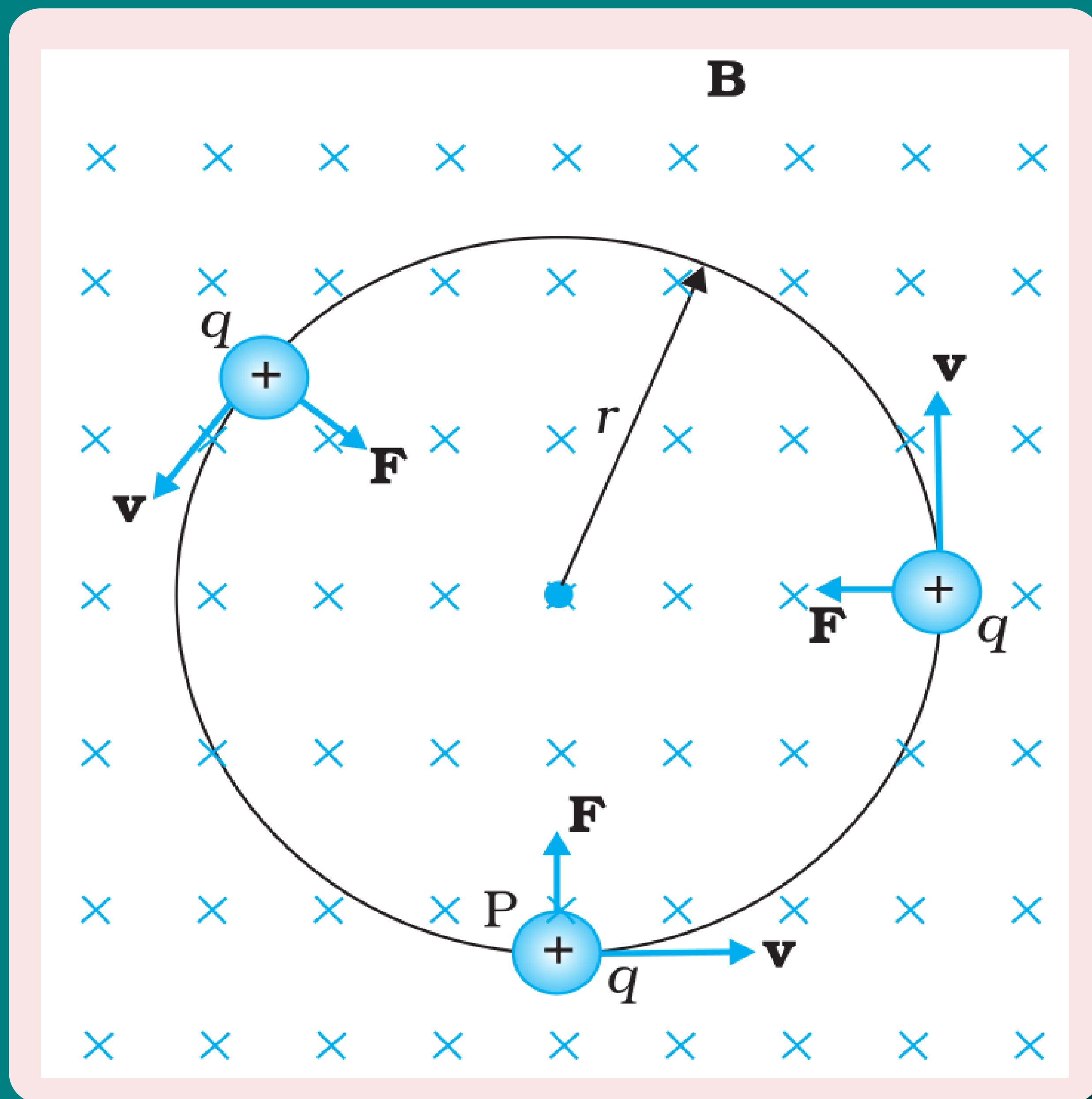


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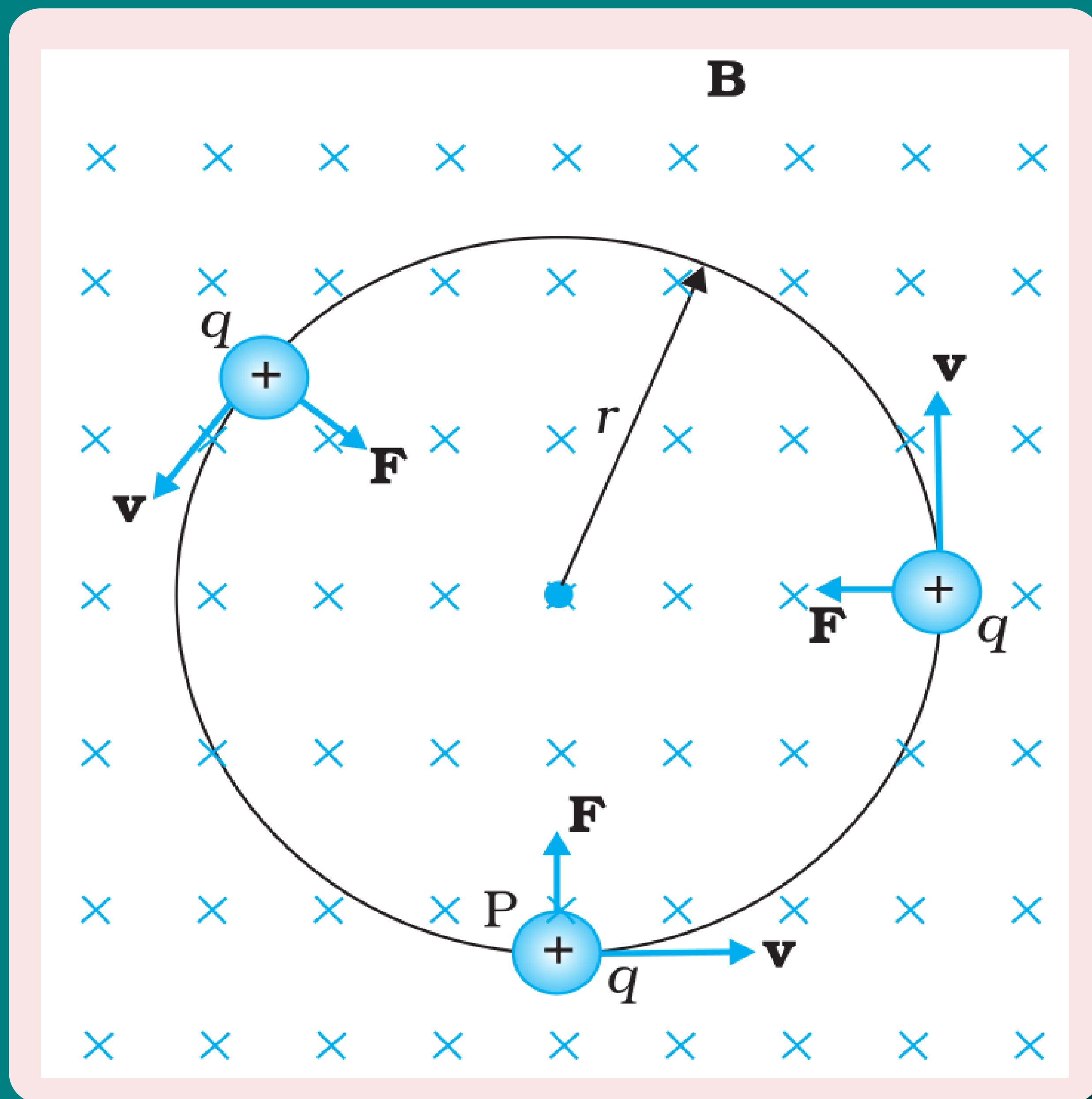
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Case 2: If velocity $\vec{v}$ has a component along $\vec{B}$

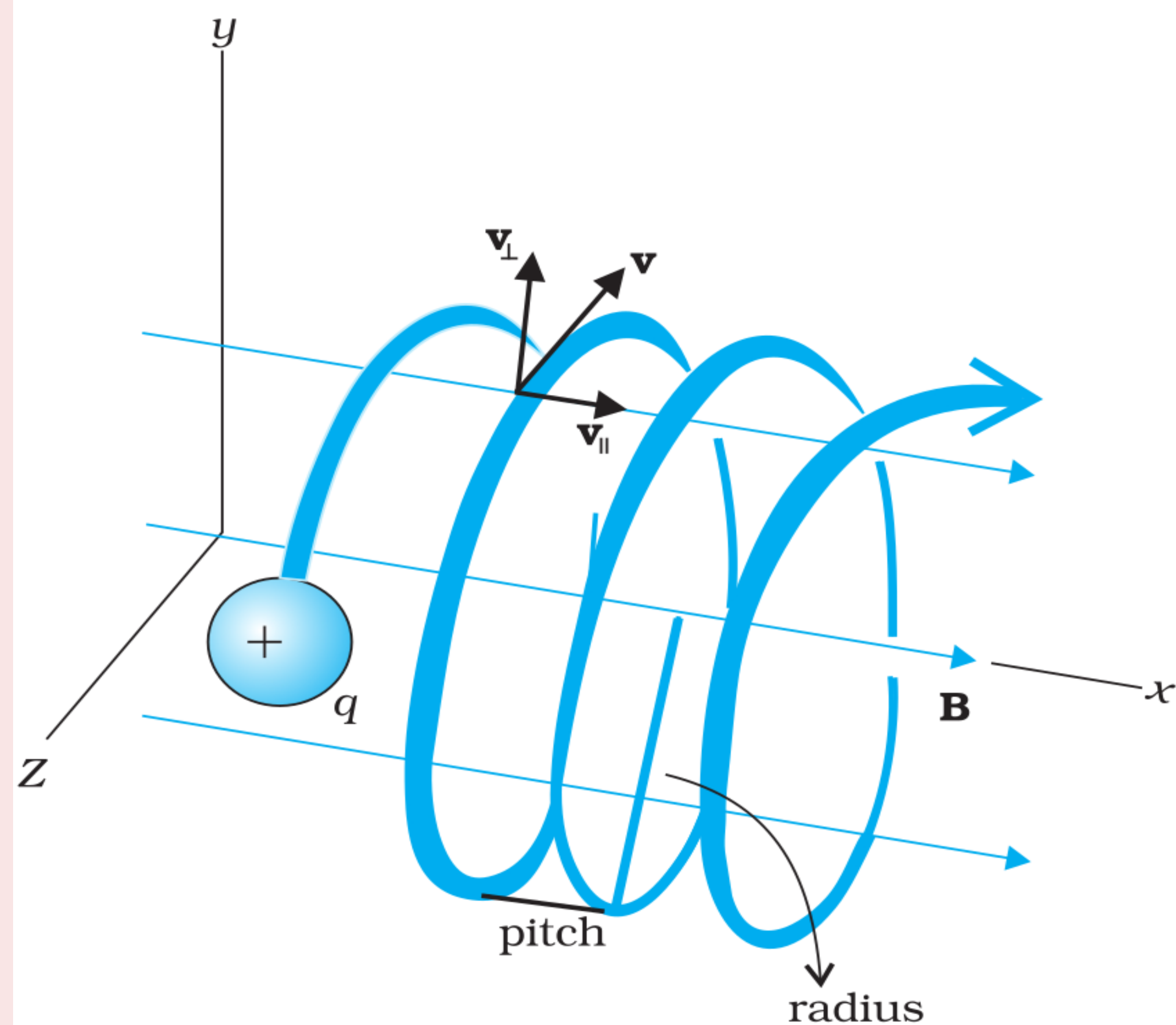


FIGURE 4.6 Helical motion

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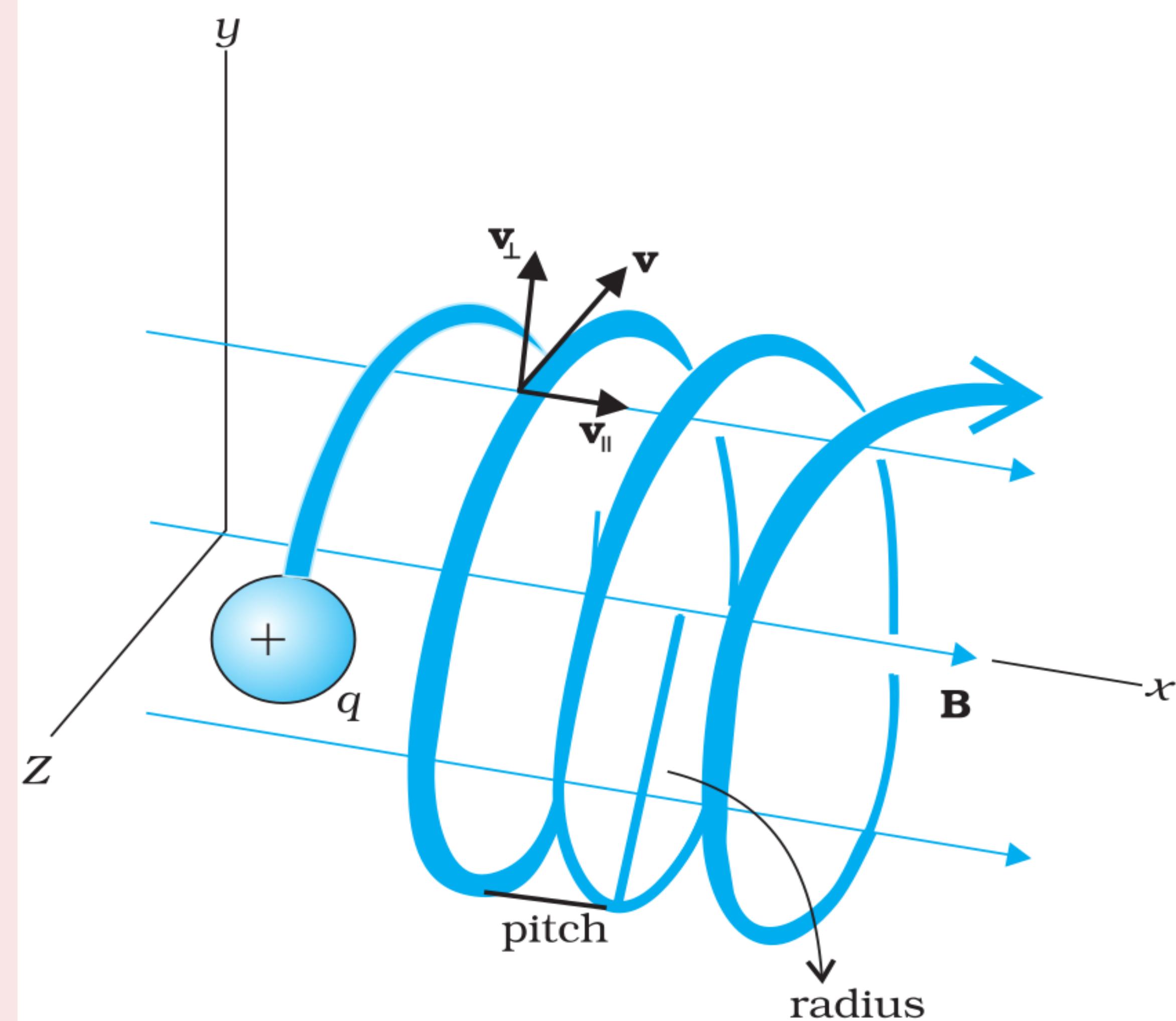


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Case 2: If velocity $\vec{v}$ has a component along $\vec{B}$

- In this case the **component of velocity along $\vec{B}$** (*i.e.*, $v_{\parallel}$) make the particle move along the field and the **path of the particle** would be a **helical**.

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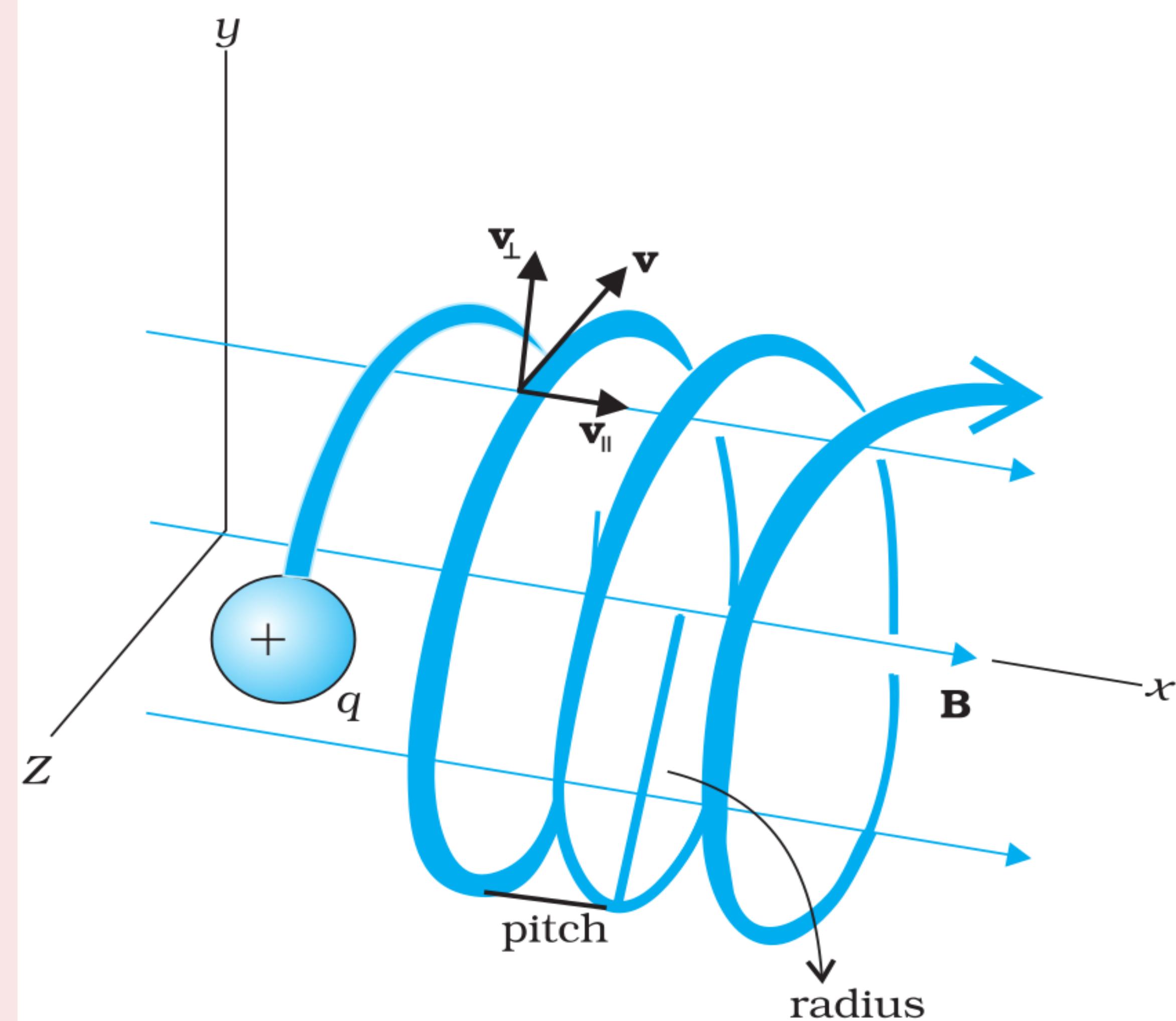


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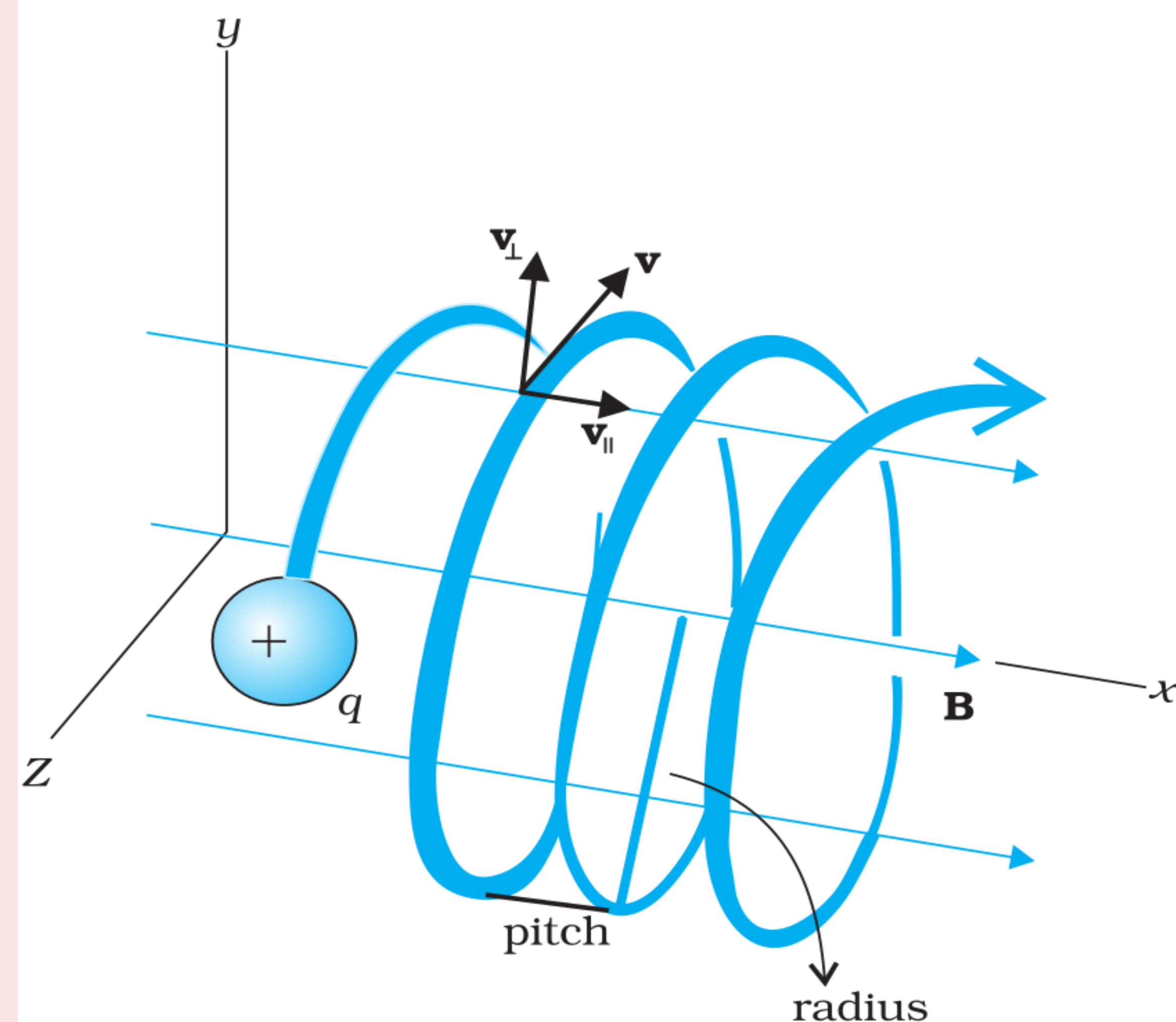


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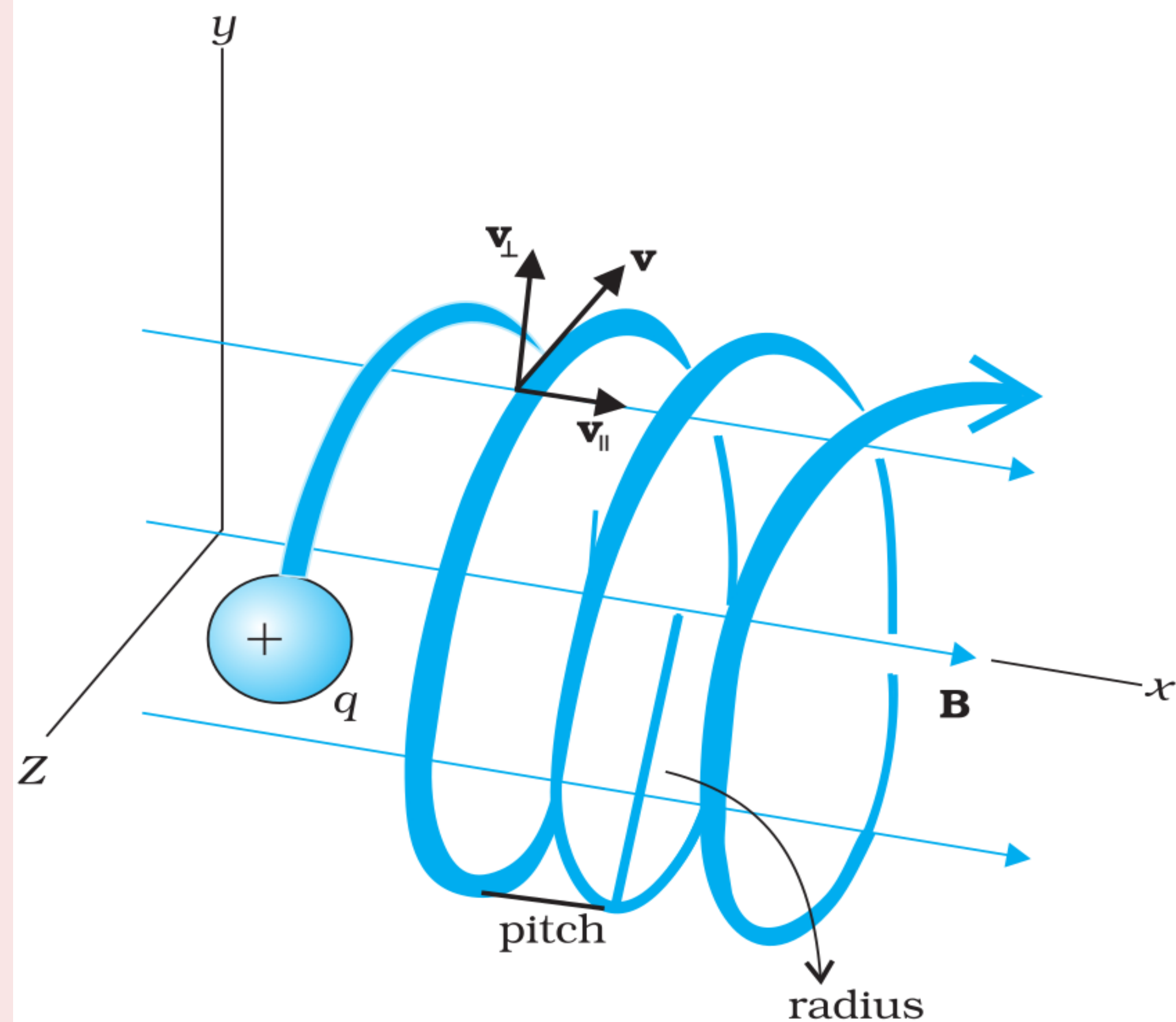


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Example 4.3

What is the radius of the path of an electron (mass $9 \times 10^{-31} \text{ kg}$ and charge $1.6 \times 10^{-19} \text{ C}$) moving at a speed of $3 \times 10^7 \text{ m/s}$ in a magnetic field of $6 \times 10^{-4} \text{ T}$ perpendicular to it? What is its frequency? Calculate its energy in keV. ($1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$).

Exercise 4.11

In a chamber, a uniform magnetic field of 6.5 G ($1 \text{ G} = 10^{-4} \text{ T}$) is maintained. An electron is shot into the field with a speed of $4.8 \times 10^6 \text{ ms}^{-1}$ normal to the field. Explain why the path of the electron is a circle. Determine the radius of the circular orbit. ($e = 1.5 \times 10^{-19} \text{ C}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$)

Exercise 4.12

In Exercise 4.11 obtain the frequency of revolution of the electron in its circular orbit. Does the answer depend on the speed of the electron? Explain.

Reference Example

A proton enters a uniform magnetic field of $1.0 \times 10^{-4} \text{ T}$ with a speed of $5 \times 10^5 \text{ m/s}$. At what angle must the magnetic field be from the velocity so that the pitch of the resulting helical motion is equal to the radius of the helix?